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A FAST TEMPLATE PERIODOGRAM FOR DETECTING NON-SINUSOIDAL FIXED-SHAPE SIGNALS IN IRREGULARLY SAMPLED TIME SERIES

J. Hoffman[1], J. Vanderplas[2], J. D. Hartman[3], G. Á Bakos[3,4]

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ABSTRACT

Astrophysical time series often contain periodic signals. The large and growing volume of time series data from photometric surveys demands computationally efficient methods for detecting and characterizing such signals. The most efficient algorithms available for this purpose are those that exploit the $\mathcal{O}(N \log N)$ scaling of the Fast Fourier Transform (FFT). However, these methods are not optimal for non-sinusoidal signal shapes. Template fits (or periodic matched filters) optimize sensitivity for *a priori* known signal shapes—for example, the characteristic lightcurves of RR Lyrae, Cepheids, δ Scuti variables, eclipsing binaries, and exoplanetary transits—but at a significant computational cost. Current implementations of template periodograms scale as $\mathcal{O}(N_f N_{\rm obs})$, where N_f is the number of trial frequencies and $N_{\rm obs}$ is the number of lightcurve observations, and due to non-convexity, they do not guarantee the best fit at each trial frequency, which can lead to spurious results. In this work, we present a non-linear extension of the Lomb-Scargle periodogram to obtain a template-fitting algorithm that is both accurate (globally optimal solutions are obtained except in pathological cases) and computationally efficient (scaling as $\mathcal{O}(N_f \log N_f)$ for a given template). The non-linear optimization of the template fit at each frequency is recast as a polynomial zero-finding problem, where the coefficients of the polynomial can be computed efficiently with the non-equispaced fast Fourier transform. We show that our method, which uses truncated Fourier series to approximate templates, is an order of magnitude faster than existing algorithms for small problems ($N \lesssim 10$ observations) and 2 orders of magnitude faster for long base-line time series with $N_{\rm obs} \gtrsim 10^4$ observations. Signals with very sharp features (e.g. limb-darkened transits with brief ingress, sharp eclipses) require moderate-to-high harmonic orders ($H \gtrsim 20$) to be reproduced at the 99% L^2 level (Table 1); for such signals, transit-tailored methods such as the Box Least-Squares periodogram may be more efficient. An open-source implementation of the fast template periodogram is freely available online.[a]

KEYWORDS

methods: data analysis – methods: statistical – techniques: photometric – stars: variables: general – stars: variables: RR Lyrae

1. INTRODUCTION

Astronomical systems exhibit a wide range of time-dependent variability. By measuring and characterizing this variability, astronomers are able to infer a variety of important astrophysical properties about the underlying system.

Periodic signals in noisy astronomical timeseries can be detected with a number of techniques, including Gaussian process regression (Foreman-Mackey et al. 2017; Rasmussen & Williams 2005), least-squares spectral analysis (Lomb 1976; Scargle 1982; Barning 1963; Vaníček 1971), and information-theoretic methods (Graham et al. 2013a; Huijse et al. 2012; Cincotta et al. 1995). For an empirical comparison of some of these techniques applied to several astronomical survey datasets, see Graham et al. (2013b).

For stationary periodic signals, least-squares spectral analysis — also known as the Lomb-Scargle (LS) periodogram (Lomb 1976; Scargle 1982; Barning 1963; Vaníček 1971) — is perhaps the most sensitive and computationally efficient method of detection. The LS periodogram can be made to scale as $\mathcal{O}(N_f \log N_f)$, where N_f is the number of trial frequencies, by utilizing the non-equispaced fast Fourier transform (Keiner et al. 2009; Dutt & Rokhlin 1993, NFFT) to evaluate frequency-dependent sums, or by "extirpolating" irregularly spaced observations to a regular grid with Lagrange polynomials (Press & Rybicki 1989).

The LS periodogram fits the following model to a set of observations:

$$\hat{y}_{\rm LS}(t; \theta, \omega) = \theta_0 \cos \omega t + \theta_1 \sin \omega t. \tag{1}$$

where $\omega = 2\pi/P$ is the (angular) frequency of the underlying signal, and θ_0 and θ_1 are the amplitudes of the

johnh2o2@gmail.com
[1] Xaxis, LLC, 3 World Trade Center, 175 Greenwich Street, 30th Floor, New York, NY 10007
[2] eScience Institute, University of Washington, Seattle, WA 98195
[3] Department of Astrophysical Sciences, Princeton University, Princeton NJ 08540
[4] Institute for Advanced Study, 1 Einstein Drive, Princeton, NJ

[a] https://github.com/PrincetonUniversity/FastTemplatePeriodogram

signal. When the data y_i is composed of a sinusoidal component and a white noise component (i.e., when the measurement uncertainties are uncorrelated and Gaussian), the LS periodogram provides a maximum likelihood estimate for the model parameters (ω, θ_0, and θ_1).

The LS "power" $P_{LS}(\omega)$ has several definitions in the literature (Zechmeister & Kürster 2009), but we adopt the following definition throughout the paper:

$$P(\omega) = \frac{V_0 - V(\omega)}{V_0} \tag{2}$$

where V_0 is the weighted sum of squared residuals for a constant fit:

$$V_0 = \sum_i w_i (y_i - \bar{y})^2 \tag{3}$$

with $\bar{y} = \sum_i w_i y_i$ the weighted mean of the observations and $w_i \propto \sigma_i^{-2}$ the normalised weights for each observation ($\sum_i w_i = 1$), and $V(\omega)$ is the weighted sum of squared residuals for the best-fit model $\hat{y}$ at a given trial frequency $\omega = 2\pi/P$ where P is the period:

$$V(\omega) = \min_\theta \sum_i w_i (y_i - \hat{y}(t_i; \omega, \theta))^2. \tag{4}$$

We use the symbol V rather than χ^2 to follow the convention common in the linear least-squares literature: with $\sum_i w_i = 1$ the quantity $\sum_i w_i (y_i - \hat{y}_i)^2$ is the weighted sum of squared residuals but is not strictly χ^2-distributed, so the χ^2 symbol can be misleading.

1.1. *Maximum-likelihood interpretation*

Although the periodogram in Equation 2 is most commonly introduced as a least-squares fit, it has a direct probabilistic reading. Under the standard noise assumptions made throughout this paper – uncorrelated, zero-mean, Gaussian uncertainties $y_i \sim \mathcal{N}(\hat{y}(t_i; \omega, \theta), \sigma_i^2)$ – the likelihood of the data given the model parameters is

$$p(\mathcal{D}|\omega, \theta) = \prod_{i=1}^{N_{\rm obs}} \mathcal{N}(\hat{y}(t_i; \omega, \theta), \sigma_i^2), \tag{5}$$

where we use $\mathcal{D} = \{(t_i, y_i, \sigma_i) \,|\, 0 < i \leq N_{\rm obs}\}$ as shorthand for the lightcurve observations (avoiding the symbol X, which is conventionally reserved for the design matrix). The log-likelihood is

$$\log p(\mathcal{D}|\omega, \theta) = -\frac{1}{2} \sum_{i=1}^{N_{\rm obs}} \left(\frac{y_i - \hat{y}(t_i; \omega, \theta)}{\sigma_i} \right)^2 + \text{const.} \tag{6}$$

$$= -\frac{W}{2} V(\theta, \omega) + \text{const.}, \tag{7}$$

where $W = \sum_i \sigma_i^{-2}$ and $V(\theta, \omega) \equiv \sum_i w_i (y_i - \hat{y}(t_i; \omega, \theta))^2$ is the weighted sum of squared residuals written with the normalised weights $w_i = \sigma_i^{-2}/W$ ($\sum_i w_i = 1$). Hence

$$V(\omega) = \min_\theta \sum_i w_i (y_i - \hat{y}(t_i; \omega, \theta))^2 \tag{8}$$

$$= -\frac{2}{W} \max_\theta \log p(\mathcal{D}|\omega, \theta) + \text{const.}, \tag{9}$$

and therefore, since $P(\omega) = 1 - V(\omega)/V_0$,

$$P(\omega) = \frac{2}{W V_0} \max_\theta \log p(\mathcal{D}|\omega, \theta) + \text{const.} \tag{10}$$

The Lomb-Scargle power is therefore a linear, monotonically increasing transformation of the profile log-likelihood – that is, of the log-likelihood maximised over the non-frequency parameters θ at each trial ω. Choosing the frequency that maximises the periodogram is equivalent to selecting the maximum-likelihood frequency estimate under the assumed noise model.

It is tempting to recast this as a maximum-a-posteriori (MAP) statement under an implicit uniform prior on θ, but we deliberately avoid that framing for two reasons. First, a uniform prior over an unbounded domain is improper; recovering the periodogram as a strict MAP quantity would require a piecewise prior on (θ_0, θ_1) that vanishes outside some explicitly chosen bounding box, which we do not impose. Second, even when bounded, a uniform prior on (θ_0, θ_1) implicitly imposes a linear prior on the sinusoid amplitude $A \equiv \sqrt{\theta_0^2 + \theta_1^2}$, since the prior volume element in the (θ_0, θ_1) plane is $\mathrm{d}\theta_0 \, \mathrm{d}\theta_1 = A \, \mathrm{d}A \, \mathrm{d}\phi$. A uniform prior on the Cartesian coefficients therefore biases the inference towards large amplitudes. The same pitfall is recognised in exoplanet eccentricity inference, where the parameterisation $(e \cos \varpi, \, e \sin \varpi)$ – introduced for sampling efficiency by Ford (2006) – implicitly imposes a linear prior on the eccentricity e that has since been corrected for or replaced (Eastman et al. 2013). The maximum-likelihood reading invoked here makes none of these implicit prior assumptions, and we adopt it throughout.

A genuinely Bayesian treatment of the LS periodogram, in which the non-frequency parameters are marginalised rather than profiled, was developed by Mortier et al. (2015a,b). Their periodogram power is

$$P_{\rm Bayes}(\omega) \propto p(\omega|\mathcal{D}) = \int p(\omega, \theta|\mathcal{D}) \, \mathrm{d}\theta,$$

evaluated under the same uncorrelated-Gaussian noise model but with explicit, bounded priors on the non-frequency parameters. Mortier et al. (2015a) focus on the classical periodogram model of Lomb (1976) and Scargle (1982) together with the constant-offset extension of Zechmeister & Kürster (2009); the same marginalisation strategy could in principle be applied to the template-based formulation developed in this paper.

As remarked in (VanderPlas 2018), the probabilistic interpretation of a periodogram is only as good as the noise model underlying it; in practice the working assumptions (uncorrelated, Gaussian uncertainties, no systematics) are often broken in real astronomical timeseries, and the maximum-likelihood reading above should be understood with the same caveats.

1.2. *Extending Lomb-Scargle*

The LS periodogram has numerous extensions to account for, e.g., biased estimates of the mean brightness (Zechmeister & Kürster 2009), non-sinusoidal signals (Schwarzenberg-Czerny 1996; Bretthorst 2003; Palmer 2009; Baluev 2009, 2013c; Wheeler & Kipping 2019), multiple periodicities (Baluev 2013a,b), multi-band observations (VanderPlas & Ivezić 2015), and to mitigate overfitting of more flexible models via regularization (VanderPlas & Ivezić 2015). Baluev (2008) provided the framework for improving false alarm statistics with extreme value theory, and this was generalized further in ?. For a more detailed review of the LS periodogram and its extensions, see VanderPlas (2018).

The multi-harmonic LS periodogram (Bretthorst et al. 1988; Schwarzenberg-Czerny 1996; Palmer 2009, MHGLS), provides a more flexible model by adding harmonic components to the fit:

$$\hat{y}_{\mathrm{MHLS}}(t;\theta,\omega) = \theta_0 + \sum_{n=1}^{H} \theta_{2n}\cos n\omega t + \theta_{2n-1}\sin n\omega t. \tag{11}$$

Additional harmonics are important for modeling signals that, while stationary and periodic, are non-sinusoidal (e.g. RR Lyrae, eclipsing binaries, etc.).

However, the MHLS periodogram contains $2H + 1$ free parameters while the original LS periodogram contains only 2 (3 if the mean brightness is considered a free parameter). Including higher order harmonics adds model complexity, which can degrade the sensitivity of the MHLS periodogram to sinusoidal or approximately sinusoidal signals.

Tikhonov regularisation (also known as L_2 regularisation) is one tool for mitigating overfitting of the higher-order harmonics. Adding an L_2 penalty on the Fourier amplitudes is equivalent to imposing a zero-mean Gaussian prior on those amplitudes; in the limit of infinite prior width, this prior reduces to the implicit (improper) uniform prior used in the unregularised fit. The choice of prior width is therefore a subjective modelling decision rather than a bias in the statistical sense, and its merits should be weighed against the corresponding decrease in model variance (VanderPlas & Ivezić 2015).

1.3. *Computational scaling*

The LS periodogram naively scales as $\mathcal{O}(N_f N_{\mathrm{obs}})$ where N_f is the number of trial frequencies and N_{obs} is the number of observations. However, the limiting computations of the LS periodogram involve sums of trigonometric functions over the observations. When the observations are regularly sampled, the fast Fourier transform (FFT) (Cooley & Tukey 1965) can evaluate such sums efficiently and the LS periodogram scales as $\mathcal{O}(N_f \log N_f)$.

When the data is not regularly sampled, as is the case for most astronomical time series, the LS periodogram can be evaluated quickly in one of two popular ways. The first, by Press & Rybicki (1989) involves"extirpolating" irregularly sampled data onto a regularly sampled mesh, and then performing FFTs to evaluate the necessary sums. The second, as pointed out in Leroy (2012), is to use the non-equispaced FFT (Keiner et al. 2009; Dutt & Rokhlin 1993, NFFT) to evaluate the sums; this provides roughly an order of magnitude speedup over the Press & Rybicki (1989) algorithm, and both algorithms scale as $\mathcal{O}(N_f \log N_f)$ (Leroy 2012).

There is a growing population of alternative methods for detecting periodic signals in astrophysical data. Some of these methods can reliably outperform the LS periodogram, especially for non-sinusoidal signal shapes (see Graham et al. (2013b) for a recent empirical review of period finding algorithms). However, a key advantage the LS periodogram and its extensions is speed. Virtually all other "phase-folding" methods scale as $\mathcal{O}(N_{\mathrm{obs}} \times N_f)$, where N_{obs} is the number of observations and N_f is the number of trial frequencies, while the Lomb-Scargle periodogram scales as $\mathcal{O}(N_f \log N_f)$. The virtual independence of Lomb-Scargle's computation time with respect to the number of observations (assuming $N_f \gtrsim N_{\mathrm{obs}}$) is especially valuable for lightcurves with $N_{\mathrm{obs}} \gg \log N_f \sim 50$.

Algorithmic efficiency will become increasingly important as the volume of data produced by astronomical observatories continues to grow larger. The HATNet survey (Bakos et al. 2004), for example, has already made $\mathcal{O}(10^4)$ observations of $\mathcal{O}(10^6-10^7)$ stars. The Gaia telescope (Gaia Collaboration et al. 2016) is set to produce $\mathcal{O}(10-100)$ observations of $\mathcal{O}(10^9)$ stars. The Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST; LSST Science Collaboration et al. 2009), which began science operations in 2025, is expected to make $\mathcal{O}(10^2-10^3)$ observations of $\mathcal{O}(10^{10})$ stars over its ten-year survey.

1.4. *Template periodograms*

When the shape of a stationary periodic signal is known *a priori*, then the number of degrees of freedom is the same as the original LS periodogram (with a floating mean component):

$$\hat{y}(t;\theta,\omega) = \theta_0 + \theta_1 \mathbf{M}(\omega t - \theta_2), \tag{12}$$

where $\mathbf{M} : [0, 2\pi) \to \mathbb{R}$ is a predefined periodic template (i.e., a function from the unit circle to the real line). We refer to the periodogram corresponding to this model as the "template periodogram."

As is the case for the LS periodogram, the template periodogram is equivalent to a maximum-likelihood estimate of the model parameters under the assumption that measurement uncertainties are Gaussian and uncorrelated (i.e. white noise).

This paper develops new extensions of least-squares spectral analysis for arbitrary signal shapes. For non-periodic signals this method is known as matched filter analysis, and can be extended to search for periodic signals by, e.g., phase folding the data at different trial periods.

An analysis by Sesar et al. (2016) found that template fitting significantly improved period and amplitude estimation for RR Lyrae in Pan-STARRS DR1 photometry (Chambers et al. 2016). Since the signal shapes for RR Lyrae in various bandpasses are known *a priori* (see Sesar et al. (2010)), template fitting provides an optimal estimate of amplitude and period, given that the object is indeed an RR Lyrae star well modeled by at least one of the templates. Templates were especially crucial for Pan-STARRS data, since there are typically only 35 observations per source over 5 bands (Hernitschek et al.

2016), not enough to obtain accurate amplitudes empirically by phase-folding. By including domain knowledge (i.e. knowledge of what RR Lyrae lightcurves look like), template fitting allows for accurate inferences of amplitude even for undersampled lightcurves.

However, the improved accuracy comes at substantial computational cost: the template fitting procedure took 30 minutes per CPU per object, and Sesar et al. (2016) were forced to limit the number of fitted lightcurves ($\lesssim$ 1000) in order to keep the computational costs to a reasonable level. Several cuts were made before the template fitting step to reduce the more than 1 million Pan-STARRS DR1 objects to a small enough number, and each of these steps removes a small portion of RR Lyrae from the sample. Though this number was reported by Sesar et al. (2016) to be small ($\lesssim 2\%$), it may be possible to further improve the completeness of the final sample by applying template fits to a larger number of objects, which would require either more computational resources, more time, or, ideally, a more efficient template fitting procedure.

We note that the fast template periodogram (FTP), like the LS periodogram, assumes that the reported photometric uncertainties σ_i are strictly positive and fully describe the noise. In practice, real photometric data frequently exhibit additional intrinsic scatter (or "equation error") beyond the reported σ_i, so that the model–data residual variance exceeds σ_i^2. In the least-squares context, the FITEXY algorithm (Press et al. 1992, *Numerical Recipes*; see also Press & Rybicki 1989) and the BCES estimators of Akritas & Bershady (1996) accommodate heteroscedastic measurement errors together with a homoscedastic equation-error term, and Kelly (2007) shows that a hierarchical-Bayesian, likelihood-based treatment delivers less bias and higher accuracy than these least-squares approximations (with reference implementations available as `LINMIX_ERR` in IDL, `linmix` in Python, and `lrgs` in R). Extending the FTP to jointly infer an unknown intrinsic scatter would require a likelihood-based generalization beyond the closed-form least-squares structure exploited here and is out of scope for the present paper. As a first-order practical correction, users with evidence of underestimated uncertainties can inflate each σ_i by a jitter term added in quadrature ($\sigma_i^2 \to \sigma_i^2 + s^2$), as is standard practice in radial-velocity and *TESS* analyses; the FTP machinery is unchanged.

The paper is organized as follows. Section 2 poses the problem of template fitting in the language of least squares spectral analysis and derives the fast template periodogram. Section 3 describes a freely available implementation of the new template periodogram. Section 4 summarizes our results, addresses caveats, and discusses possible avenues for improving the efficiency of the current algorithm.

2. DERIVATIONS

We define a template $\mathbf{M}$

$$\mathbf{M} : [0, 2\pi) \to \mathbb{R}, \tag{13}$$

as a mapping between the unit interval and the set of real numbers. We restrict our discussion to sufficiently smooth templates such that $\mathbf{M}$ can be adequately described by a truncated Fourier series

$$\hat{\mathbf{M}}(\omega t|H) = \sum_{n=1}^{H} (c_n \cos n\omega t + s_n \sin n\omega t) \tag{14}$$

for some finite $H > 0$.

That the c_n and s_n values are *fixed* (i.e., they define the template) is the crucial difference between the template periodogram and the multi-harmonic Lomb-Scargle (Palmer 2009; Bretthorst et al. 1988), where c_n and s_n are *free parameters*.

We now construct a periodogram for this template. The periodogram assumes that an observed time series $S = \{(t_i, y_i, \sigma_i)\}_{i=1}^{N}$ can be modeled by a scaled, transposed template that repeats with period $2\pi/\omega$, i.e.

$$y_i \approx \hat{y}(\omega t_i; \theta, \mathbf{M}) = \theta_1 \mathbf{M}(\omega t_i - \theta_2) + \theta_3, \tag{15}$$

where $\theta = (\theta_1, \theta_2, \theta_3) \in \mathbb{R}^3$ is a set of model parameters.

The optimal parameters are the location of a local minimum of the (weighted) sum of squared residuals,

$$V(\theta, S) \equiv \sum_i w_i (y_i - \hat{y}(\omega t_i; \theta))^2, \tag{16}$$

and thus the following condition must hold for all three model parameters at the optimal solution $\theta = \theta_{\rm opt}$:

$$\left.\frac{\partial V}{\partial \theta_j}\right|_{\theta=\theta_{\rm opt}} = 0 \quad \forall \theta_j \in \theta. \tag{17}$$

Note that we have implicitly assumed $V(\theta, S)$ is a C^1 differentiable function of θ, which requires that $\mathbf{M}$ is a C^1 differentiable function. Though this assumption could be violated if we considered a more complete set of templates, (e.g. a box function), our restriction to truncated Fourier series ensures C^1 differentiability.

Note that we also implicitly assume $\sigma_i > 0$ for all i and we will later assume that the variance of the observations y_i is non-zero. If there are no measurement errors, i.e. $\sigma_i = 0$ for all i, then uniform weights (setting $\sigma_i = 1$) should be used. If the variance of the observations y is zero, the periodogram (as defined in Equation 2) is undefined for all frequencies. We do not consider the case where $\sigma_i = 0$ for some observations i and $\sigma_j > 0$ for some observations j.

We can derive a system of equations for $\theta_{\rm opt}$ from Equation 17. The fast template periodogram reduces this three-parameter non-linear minimisation to a polynomial root-finding problem in three steps; the qualitative outline is given here, with the explicit algebra collected in Appendix A.

Step 1: closed form for (θ_1, θ_3) at fixed θ_2. The model $\hat{y}$ is linear in θ_1 and θ_3 once the phase shift θ_2 is held fixed, so the conditions $\partial V/\partial\theta_1 = \partial V/\partial\theta_3 = 0$ form a 2×2 linear system with closed-form solution (Appendix A, Equation A11):

$$\theta_1 = \frac{YM}{MM}, \qquad \theta_3 = \bar{y} - \overline{M}\left(\frac{YM}{MM}\right), \tag{18}$$

where the weighted statistics

$$\overline{M} \equiv \langle \mathbf{M}_{\theta_2} \rangle\,, \qquad \bar{y} \equiv \langle y \rangle\,,$$
$$MM \equiv \mathrm{Var}(\mathbf{M}_{\theta_2}), \qquad YM \equiv \mathrm{Cov}(\mathbf{M}_{\theta_2}, y),$$
$$YY \equiv \mathrm{Var}(y) \tag{19}$$

are all implicit functions of θ_2 through the phase-shifted template $\mathbf{M}_{\theta_2} \equiv \mathbf{M}(\omega t - \theta_2)$, and the weighted-average operator $\langle\cdot\rangle$, covariance Cov, and variance Var are defined in Appendix A. Substituting Equation 18 back into V and using $V_0 = YY$, the periodogram $P(\omega) \equiv 1 - V/V_0$ takes the compact form

$$P(\omega) = \frac{[YM(\theta_2)]^2}{YY \cdot MM(\theta_2)}. \tag{20}$$

The intermediate algebra is given in Appendix A.

Step 2: reduce to a one-dimensional optimisation over θ_2. Maximising Equation 20 with respect to the remaining non-linear parameter θ_2 gives

$$\begin{aligned}\partial_{\theta_2} P = 0 \;&=\; \frac{YM}{YY \cdot MM}\left(2\partial_{\theta_2}(YM) - \frac{YM}{MM}\partial_{\theta_2}(MM)\right) \\ &=\; 2\,MM\,\partial_{\theta_2}(YM) - YM\,\partial_{\theta_2}(MM).\end{aligned} \tag{21}$$

This transcendental condition on the single parameter θ_2 is what existing non-linear template fitters must solve numerically at every trial frequency. Satisfying Equation 21 is necessary but not *sufficient*: the value of $P(\omega)$ must be evaluated at each root, and the global maximum selected from this set.

Step 3: recast as polynomial root-finding. The crux of the FTP is the change of variable $\psi \equiv e^{i\theta_2}$ together with the rescaling $YM' \equiv \psi^H\,YM$ and $MM' \equiv \psi^{2H}\,MM$. Because the phase-shifted template (Equation 14) enters YM and MM only through factors of $e^{\pm in\theta_2}$ with $n \le H$, the rescaled quantities YM' and MM' are polynomials in ψ (of degree $2H$ and $4H$, respectively; Appendix A). A short calculation (Appendix A) shows that the non-linear condition Equation 21 is equivalent to

$$0 \;=\; 2\,MM'\,\partial(YM') \;-\; YM'\,\partial(MM'), \tag{22}$$

a polynomial of degree $6H-1$ in ψ. The coefficients of this polynomial are linear combinations of the weighted trigonometric sums

$$CC_{nm} \equiv \mathrm{Cov}(\cos n\omega t, \cos m\omega t), \tag{23}$$
$$CS_{nm} \equiv \mathrm{Cov}(\cos n\omega t, \sin m\omega t), \tag{24}$$
$$SS_{nm} \equiv \mathrm{Cov}(\sin n\omega t, \sin m\omega t), \tag{25}$$
$$YC_n \equiv \langle (y - \bar{y}) \cos n\omega t \rangle\,, \tag{26}$$
$$YS_n \equiv \langle (y - \bar{y}) \sin n\omega t \rangle\,, \tag{27}$$

each of which can be evaluated at *all* trial frequencies ω simultaneously in $\mathcal{O}(HN_f \log HN_f)$ time using the non-equispaced fast Fourier transform. The explicit assembly of the polynomial coefficients from these sums—and the algebraic identity that converts Equation 21 into Equation 22—is given in Appendix A (Equations A19–A26).

We solve for the roots of Equation 22 using the `numpy.polynomial.polyroots` function, which returns the eigenvalues of the polynomial's companion matrix. Multiplying through by powers of ψ in deriving Equation 22 introduces spurious roots, and ψ must lie on the unit circle ($|\psi| = 1$ for real θ_2). We therefore scale each root by its modulus to enforce the unit-circle constraint; iterative schemes such as Newton's method could be used for greater robustness, but the maximum periodogram error of the present implementation is below 10^{-7} across the H values tested (Section 2.4).

Summary of the algorithm. We have derived an explicit, non-linear system of equations to solve for the optimal model parameters at each trial frequency. Concretely, for each ω the FTP proceeds as follows:

1. Compute the weighted trigonometric sums of Equations 23–27 at all trial frequencies ω using the NFFT.

2. Assemble the coefficients of the degree-$(6H-1)$ polynomial of Equation 22 from the sums of step 1; the closed-form assembly is given in Appendix A (Equations A19–A23).

3. Find all roots of this polynomial via `numpy.polynomial.polyroots`, retaining only the roots that lie on the unit circle $\psi = e^{i\theta_2}$.

4. For each candidate θ_2, evaluate the corresponding linear-best-fit (θ_1, θ_3) via Equation 18 and the periodogram $P(\omega)$ via Equation 20.

5. Declare the largest $P(\omega)$ value the global optimum; the corresponding $(\theta_1, \theta_2, \theta_3)$ are the best-fit model parameters at that trial frequency.

2.1. *Negative amplitude solutions*

The model $\hat{y} = \theta_1 \mathbf{M}_{\theta_2} + \theta_0$ allows for $\theta_1 < 0$ solutions. In the original formulation of Lomb-Scargle and in linear extensions involving multiple harmonics, negative amplitudes translate to phase differences, since $-\cos x = \cos(x - \pi)$ and $-\sin x = \sin(x - \pi)$.

However, for non-sinusoidal templates, $\mathbf{M}$, negative amplitudes do not generally correspond to a phase difference. For example, a detached eclipsing binary template $\mathbf{M}_{\mathrm{EB}}(x)$ cannot be expressed in terms of a phase-shifted negative eclipsing binary template; i.e. $\mathbf{M}_{\mathrm{EB}} \neq -\mathbf{M}_{\mathrm{EB}}(x - \phi)$ for any $\phi \in [0, 2\pi)$.

Negative amplitude solutions found by the fast template periodogram are usually undesirable, as they may produce false positives for lightcurves that resemble flipped versions of the desired template, and allowing for $\theta_1 < 0$ solutions increases the number of effective free parameters of the model, which lowers the signal to noise, especially for weak signals.

One possible remedy for this problem is to set $P_{\mathrm{FTP}}(\omega) = 0$ if the optimal solution for θ_1 is negative, but this complicates the interpretation of P_{FTP}. Another possible remedy is, for frequencies that have a $\theta_1 < 0$ solution, to search for the optimal parameters while enforcing that $\theta_1 > 0$, e.g. via non-linear optimization, but this likely will eliminate the computational advantage of FTP over existing methods.

Thus, we allow for negative amplitude solutions in the model fit and caution the user to check that the best fit θ_1 is positive.

2.2. *Extending to multi-band observations*

2.2.1. *Multi-phase model*

As shown in VanderPlas & Ivezić (2015), the multi-phase periodogram (their $(N_{\rm base}, N_{\rm band}) = (0,1)$ periodogram), for any model can be expressed as a linear combination of single-band periodograms:

$$P^{(0,1)}(\omega) = \frac{\sum_{k=1}^{K} V_{0,k}\, P_k(\omega)}{\sum_{k=1}^{K} V_{0,k}} \tag{28}$$

where K denotes the number of bands, $V_{0,k}$ is the weighted sum of squared residuals between the data in the k-th band and its weighted mean $\langle y\rangle$, and $P_k(\omega)$ is the periodogram value of the k-th band at the trial frequency ω.

With Equation 28, the template periodogram is readily applicable to multi-band time series, which is crucial for experiments like LSST, SDSS, Pan-STARRS, and other current and future photometric surveys.

Other multi-band extensions of the template periodogram are provided in Appendix B.

2.3. *Computational requirements*

For a given number of harmonics H, the task of deriving the polynomial given in Equation 22 requires $\mathcal{O}(H^2)$ computations, and finding the roots of this polynomial requires $\mathcal{O}(H^3)$ computations. The degree of the final polynomial is $6H - 1$.

When considering N_f trial frequencies, the polynomial computation and root-finding step scales as $\mathcal{O}(H^3 N_f)$. The computation of the sums (Equations 23 – 27) scales as $\mathcal{O}(HN_f \log HN_f)$.

Therefore, the entire template periodogram scales as

$$\mathcal{O}(HN_f \log HN_f) \;+\; \mathcal{O}(H^3 N_f). \tag{29}$$

We have written the bound as the sum of two independent contributions in order to emphasize that the two terms are governed by different operations and dominate in different regimes. The first term, from the non-equispaced fast Fourier transforms used to compute the C and S sums of Equations 23–27, dominates at low H and large N_f – the regime typical of large surveys with near-sinusoidal templates. The second term, from constructing and rooting the degree-$(6H-1)$ polynomial of Equation 22 at every trial frequency, dominates at moderate-to-high H and moderate N_f – the regime typical of sharply peaked templates such as eclipsing-binary or planet-transit shapes.

However, an important consideration is that the peak width scales inversely with the number of harmonics $\delta f_{\rm peak} \propto 1/H$ and so the number of trial frequencies needed to resolve a peak increases linearly with the number of harmonics in the template.

The computational scaling factoring in an extra power of H is therefore

$$\mathcal{O}(H^2 N_f \log HN_f + H^4 N_f). \tag{30}$$

For a fixed number of harmonics H, the template periodogram scales as $\mathcal{O}(N_f \log N_f)$. However, for a constant number of trial frequencies N_f, the template algorithm scales as $\mathcal{O}(H^3)$, and computational resources alone limit H to reasonably small numbers $H \lesssim 15$ (see Figure 1).

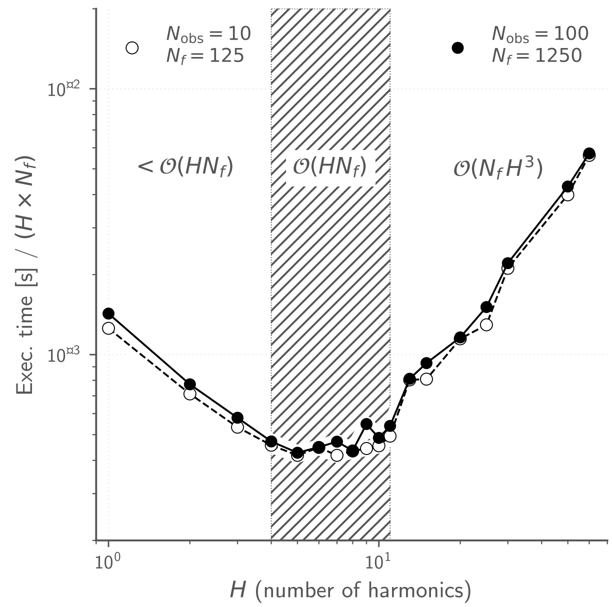

FIG. 1.— Computation time of FTP scaled by NH for different numbers of harmonics. For $H \lesssim 3$, FTP scales sublinearly in H (possibly due to a constant overhead per trial frequency, independent of H). When $3 \lesssim H \lesssim 11$, FTP scales approximately linearly in H, and when $H \gtrsim 11$ FTP approaches the $\mathcal{O}(H^3)$ scaling limit.

This practical ceiling has a direct bearing on the algorithm's scope of applicability for box-like signals such as exoplanetary transits and detached eclipsing binaries. As we calibrate in Section 2.4, the canonical analytic trapezoidal-transit and detached-EB templates considered here have second-derivative discontinuities at their ingress/egress corners, which produce a $\sim 1/n^2$ tail in the Fourier coefficients and require $H \approx 12$ to reach 95% L^2 fidelity and $H \approx 21$ to reach 99% (Table 1). FTP therefore *can* handle transit- and eclipse-shaped signals, but at the upper end of computationally tractable H values, and at a cost that grows as $H^3 N_f$. Real limb-darkened transits are smoother than the trapezoidal idealisation—they have finite, gently-curved ingress/egress and consequently Fourier coefficients that decay faster than the trapezoid's $1/n^2$ tail—so the H requirements in practice may be lower than Table 1 suggests for a pathological trapezoid. Nevertheless, for surveys targeting genuinely box-shaped signals with very sharp ingress/egress (e.g. short-cadence shallow transits or short-duration eclipses), transit-tailored methods such as the Box Least-Squares periodogram (BLS, Kovács et al. 2002) are likely to be a more efficient choice than FTP.

2.4. *Choosing the harmonic order H*

The harmonic order H is a free parameter of the algorithm. Two considerations bound it in practice: (i) the L^2 fidelity of the H-truncated Fourier series as an approximation of the true template shape, and (ii) the numerical precision of the polynomial-rooting step at large H.

To quantify (i) for representative fixed-shape signals, we evaluated the cumulative explained-variance frac-

TABLE 1
The minimum harmonic order $H_{\min}(F)$ at which the H-truncated Fourier expansion of each canonical template captures the listed fraction F of its total L^2 energy. The right-hand columns report the empirical period-recovery rate at peak-to-peak SNR = 7 on 30 independent synthetic light curves with $N_{\rm obs} = 100$ irregular samples per trial, comparing the Fast Template Periodogram at $H = H_{\min}(F{=}0.95)$ with the single-harmonic Lomb–Scargle-equivalent (FTP at H=1). Tabulated values come from the analytic Fourier expansions described in Section 2.4 and from the open-source reproduction script in the paper repository.

Template	$H_{\min}(F)$			Recovery at SNR = 7		
	90%	95%	99%	H	FTP	LS-eq.
RR Lyrae ab	2	3	8	$H = 3$	100%	100%
RR Lyrae c	2	2	3	$H = 2$	100%	100%
Planet transit	6	12	21	$H = 12$	100%	47%
Detached EB	6	12	21	$H = 12$	100%	0%
Ellipsoidal + beaming	2	2	2	$H = 2$	100%	0%

tion $F(H) = \sum_{n=1}^{H}(c_n^2 + s_n^2)/\sum_{n=1}^{\infty}(c_n^2 + s_n^2)$ for five canonical analytic templates: an asymmetric exponential RR Lyrae ab pulse (fast rise, slow decline); a skewed sinusoidal RR Lyrae c profile; a trapezoidal planet transit with linear ingress/egress; a detached eclipsing binary built from two such trapezoidal eclipses; and a Doppler-beaming + ellipsoidal-modulation profile (exactly two-harmonic by construction). Table 1 lists the smallest H that reaches each of the $F = 0.90, 0.95, 0.99$ thresholds, and Figure 2(a) shows the full $F(H)$ curves.

The five templates span an order of magnitude in their harmonic requirements. At the low end, the ellipsoidal + beaming profile is exactly band-limited at $H = 2$; the skewed-sinusoid RR Lyrae c reaches 99% at $H = 3$. The asymmetric-pulse RR Lyrae ab profile reaches 99% at $H = 8$. At the high end, the trapezoidal transit and detached-EB templates have second-derivative discontinuities at their ingress/egress corners, which gives a $\sim 1/n^2$ tail in the Fourier coefficients; both require $H \approx 12$ for 95% fidelity and $H \approx 21$ for 99%. For applications where the underlying signal has sharper features than these analytic templates (e.g. true limb-darkened transits with fast cadence), the 99% values may need to be revised upward.

The practical consequence for period detection is shown in Figure 2(b), which reports the period-recovery rate of FTP and of the LS-equivalent (FTP at H=1, i.e. a single-harmonic sinusoidal template) on synthetic noisy light curves. Each light curve consists of $N_{\rm obs} = 100$ samples drawn uniformly from a $T_{\rm obs} = 100$ day window with independent Gaussian noise scaled to the indicated peak-to-peak SNR; each periodogram is evaluated on the frequency window $[0.01, 5]$ cyc day^{-1} at 10 samples per peak, and the period is declared recovered when $|f_{\rm peak} - f_{\rm true}|/f_{\rm true} < 5 \times 10^{-3}$. Thirty independent trials are run per (template, SNR) cell. For the smooth, near-sinusoidal RR Lyrae profiles the two estimators agree at all SNRs: a single-harmonic template suffices. For the three non-sinusoidal templates (transit, EB, ellipsoidal + beaming), FTP at $H = H_{95\%}$ recovers the period in $\geq 29/30$ trials at every SNR tested, while the LS-equivalent fails completely for EB and ellipsoidal + beaming (0/30 recoveries at all SNRs) and only partially for the transit template (4/30, 14/30, and 17/30 at SNR = 3, 7, 15 respectively). This is the regime in which the H-truncation matters: harmonics beyond the first are required to match the underlying signal shape.

For the numerical-precision consideration (ii), we have measured the maximum periodogram error of the implementation in this paper against a brute-force evaluation that explicitly maximises $V(\theta, \omega)$ on a dense θ_2 grid with local refinement. Across $H \in \{1, 2, 3, 5, 7\}$, five random data realisations, and 200 trial frequencies, the maximum absolute error in $P(\omega)$ is below 10^{-7}.[5] This is well below any astrophysically relevant amplitude threshold, so polynomial-rooting error does not limit FTP at the values of H considered in Table 1.

3. IMPLEMENTATION

An open-source implementation of the Fast Template Periodogram (FTP) in Python is available.[6] Polynomial algebra is performed using the `numpy.polynomial` module (Jones et al. 2001–). The `nfft` Python module, [7] which provides a Python implementation of the non-equispaced fast Fourier transform, is used to compute the necessary sums for a particular time series.

The high-level entry point, `autopower(samples_per_peak=...)`, exposes the trial-frequency grid density as a user parameter: `samples_per_peak` is the number of trial frequencies that cover one full-width of a periodogram peak. Because the peak width scales roughly as $\delta f_{\rm peak} \propto 1/H$ (Section 2.3), the value of `samples_per_peak` must scale at least linearly with H to avoid under-sampling the peaks. In particular, the conventional default `samples_per_peak` = 10, which is a reasonable choice for near-sinusoidal ($H = 1$) templates, implies only ~ 1 trial frequency per peak at $H = 10$; an empirical inspection at $H = 7$ shows that peaks are only about three samples wide on the `samples_per_peak` = 10 grid, so the true frequency can be missed in favor of a noise peak. For the sharper templates studied in Section 2.4 – in particular the trapezoidal-transit and detached-EB shapes, which require $H_{95\%} = 12$ in Table 1 – `samples_per_peak` $\gtrsim 30$ is a more defensible default. This is the runtime/grid-density versus peak-recovery tradeoff documented in the open-source implementation, and users selecting H via Section 2.4 should adjust `samples_per_peak` accordingly.

No explicit parallelism is used anywhere in the current implementation, however certain linear algebra operations in `Scipy` use OpenMP via calls to BLAS libraries that have OpenMP enabled.

All timing tests were run on a quad-core 2.6 GHz Intel Core i7 MacBook Pro laptop (mid-2012 model) with 8GB of 1600 MHz DDR3 memory. The `Scipy` stack (version 0.18.1) was compiled with multi-threaded MKL libraries.

3.1. *Comparison with non-linear optimization*

[5] Issue #33 of the FTP repository at https://github.com/PrincetonUniversity/FastTemplatePeriodogram/issues/33 documents the test methodology and the precision floor of `numpy.polynomial.polyroots` as used by the implementation.
[6] https://github.com/PrincetonUniversity/FastTemplatePeriodogram
[7] https://github.com/jakevdp/nfft

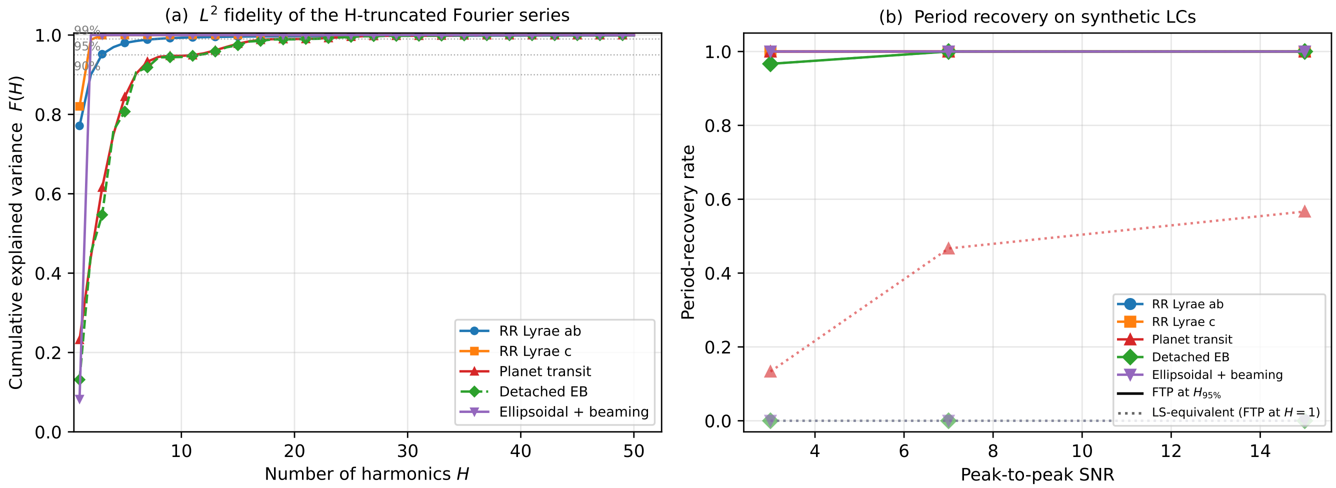

FIG. 2.— (a) Cumulative L^2 explained-variance fraction $F(H)$ of the H-truncated Fourier series as an approximation of each canonical template. Horizontal dotted guides mark $F = 0.90, 0.95, 0.99$. Detached EB and Planet transit have visually overlapping curves (both driven by sharp ingress/egress discontinuities); the dashed green line distinguishes the EB. (b) Empirical period-recovery rate as a function of peak-to-peak signal-to-noise ratio for sparse $N_{\rm obs} = 100$ synthetic light curves (30 trials per cell), comparing FTP at $H = H_{95\%}$ (solid) with the Lomb–Scargle-equivalent FTP at H=1 (dotted) for the same five canonical templates. FTP at the per-template $H_{95\%}$ recovers the period in $\geq 96\%$ of trials across the SNR range. The LS-equivalent agrees with FTP for the smooth RR Lyrae profiles (overlapping solid lines), fails completely for the EB and ellipsoidal + beaming templates (flat dotted lines at zero), and recovers only a fraction of trial transits (red dotted line, climbing from ∼13% at SNR= 3 to ∼57% at SNR= 15).

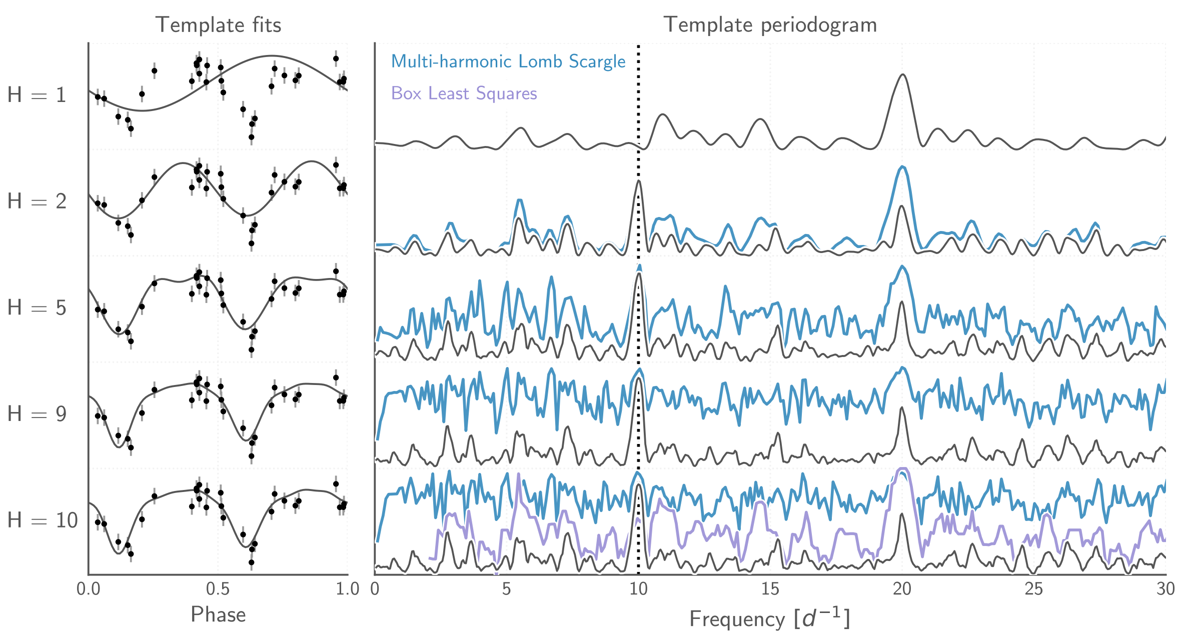

FIG. 3.— Template periodograms performed on a simulated eclipsing binary lightcurve (shown phase-folded in the left-hand plots). The top-most plot uses only one harmonic, equivalent to a Lomb-Scargle periodogram. Subsequent plots use an increasing number of harmonics, which produces a narrower and higher peak height around the correct frequency. For comparison, the multi-harmonic extension to Lomb-Scargle is plotted in blue, using the same number of harmonics as the FTP. The Box Least-Squares (Kovács et al. 2002) periodogram is shown in the final plot.

In order to evaluate the accuracy and speed of the template periodogram, we have included slower alternative solvers within the Python implementation of the FTP that employ non-linear optimization to find the best fit parameters.

Figure 3 shows example periodograms computed for simulated data, and compares the FTP to the multi-harmonic Lomb–Scargle (MHLS; Bretthorst et al. 1988; Schwarzenberg-Czerny 1996; Palmer 2009) and Box Least Squares (BLS; Kovács et al. 2002) periodograms; the remaining accuracy and timing comparisons (Figures 4 and 5) use the same simulated-data prescription. The simulated data have uniformly random observation times, with Gaussian-random, homoskedastic, uncorrelated uncertainties. An eclipsing binary template, generated by fitting a well-sampled, high signal-to-noise eclipsing binary in the HATNet dataset (2MASS J02240150+5717241, also known as BD+56° 603) with a 10-harmonic truncated Fourier series, is added to the simulated lightcurves and used as the template for the FTP search. At the correct period, the FTP yields a much stronger detection than either MHLS—which has $2H+1$ free parameters and consequently reduces the signal-to-noise of the periodogram peak relative to a fixed-shape template—or BLS, which provides only a coarse box-shaped match to the smooth EB profile.

3.1.1. *Accuracy*

For weak signals or signals folded at the incorrect trial period, the χ^2 surface in θ_2 admits up to $6H-1$ stationary points (the degree of the polynomial in Equation 22), so non-linear optimization algorithms can fail to find the global minimum and instead lock onto one of the spurious local extrema. The FTP sidesteps this entirely by enumerating all roots of Equation 22 and choosing the one that yields the largest $P(\omega)$, so it recovers the optimal solution even when the signal is weak or absent.

Figure 4 illustrates the accuracy improvement with FTP. Many solutions found via non-linear optimization are significantly suboptimal compared to the solutions found by the FTP.

Figure 5 compares FTP results obtained using the full template ($H = 10$) with those obtained using smaller numbers of harmonics. The left-most plot compares the $H = 1$ case (weighted Lomb-Scargle), which, as also demonstrated in Figure 3, illustrates the advantage of the template periodogram for known, non-sinusoidal signal shapes.

3.1.2. *Computation time*

FTP scales asymptotically as $\mathcal{O}(N_f H \log N_f H)$ with respect to the number of trial frequencies, N_f and as $\mathcal{O}(N_f H^3)$ with respect to the number of harmonics in which the template is expanded, H. For a given resolving power, however, there is an additional factor of H in each of these terms due to the number of trial frequencies necessary to resolve a periodogram peak being proportional to H. However, for reasonable cases ($N_f \lesssim 10^{120}$ when $H = 5$) the computation time is dominated by computing polynomial coefficients and root finding, both of which scale linearly in N_f.

The number of trial frequencies needed for finding astrophysical signals in a typical photometric time series

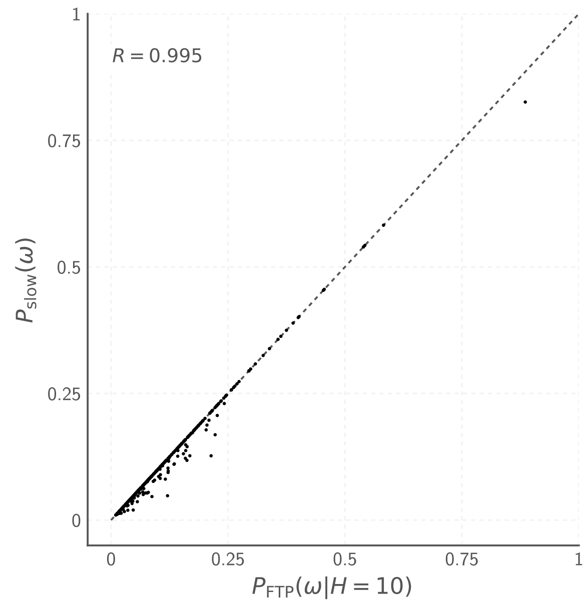

FIG. 4.— Comparing accuracy between previous methods that rely on non-linear optimization at each trial frequency with the fast template periodogram described in this paper. Each point corresponds to one realization of the simulated data (an example of which is shown in Figure 3); the axes give the *peak* periodogram value $P_{\rm FTP}(\omega \mid H{=}10)$ from the FTP and $P_{\rm slow}(\omega)$ from the non-linear "slow" template fitter, not the periodograms themselves. The annotated R is the Pearson linear-correlation coefficient between the two sets of peak values. The FTP consistently finds more optimal template fits than the non-linear solver, which does not guarantee convergence to a globally optimal solution; in realizations where the slow fitter is trapped in a local V-minimum, it reports a higher residual V and therefore a lower peak $P_{\rm slow}$ than the FTP, producing the points below the diagonal at low $P_{\rm FTP}$.

is

$$N_f = 1.75 \times 10^6 \left(\frac{H}{1}\right)\left(\frac{\text{baseline}}{10 \text{ yrs}}\right)\left(\frac{\alpha}{5}\right)\left(\frac{15 \text{ mins}}{P_{\rm min}}\right) \tag{31}$$

where α represents the "oversampling factor," $\Delta f_{\rm peak}/\Delta f$, where $\Delta f_{\rm peak} \sim 1/\text{baseline}$ is the typical width of a peak in the periodogram and Δf is the frequency spacing of the periodogram.

Extrapolating from the timing of a test case (500 observations, 5 harmonics, 15,000 trial frequencies), the summations account for approximately 5% of the computation time when $N_f \sim 10^6$. If polynomial computations and root-finding can be improved to the point where they no longer dominate the computation time, this would provide an order of magnitude speedup over the current implementation.

Figure 6 compares the timing of the FTP with that of previous methods that employ non-linear optimization. For the case when $N_f \propto N_{\rm obs}$, FTP achieves a factor of 3 speedup for even the smallest test case (15 datapoints), while for larger cases ($N \sim 10^4$) FTP offers 2-3 orders of magnitude speed improvement. For the constant baseline case, FTP is a factor of ~ 2 faster for the smallest test case and a factor of ~ 20 faster for $N_{\rm obs} \sim 10^4$. Future improvements to the FTP implementation could further

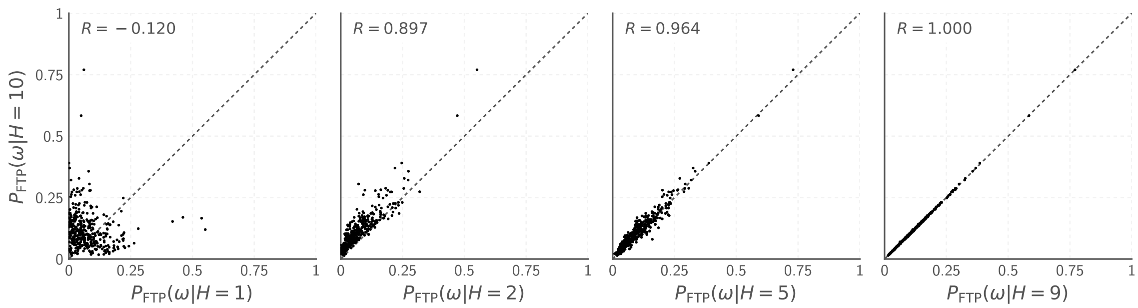

FIG. 5.— Similar to Figure 4. Here we compare the template periodogram calculated with $H = 10$ harmonics to the template periodogram using a smaller number of harmonics $H < 10$. Each point is the *peak* periodogram value from one realization; R is again the Pearson linear-correlation coefficient between the two sets of peak values. The template and data prescription match those of Figure 3.

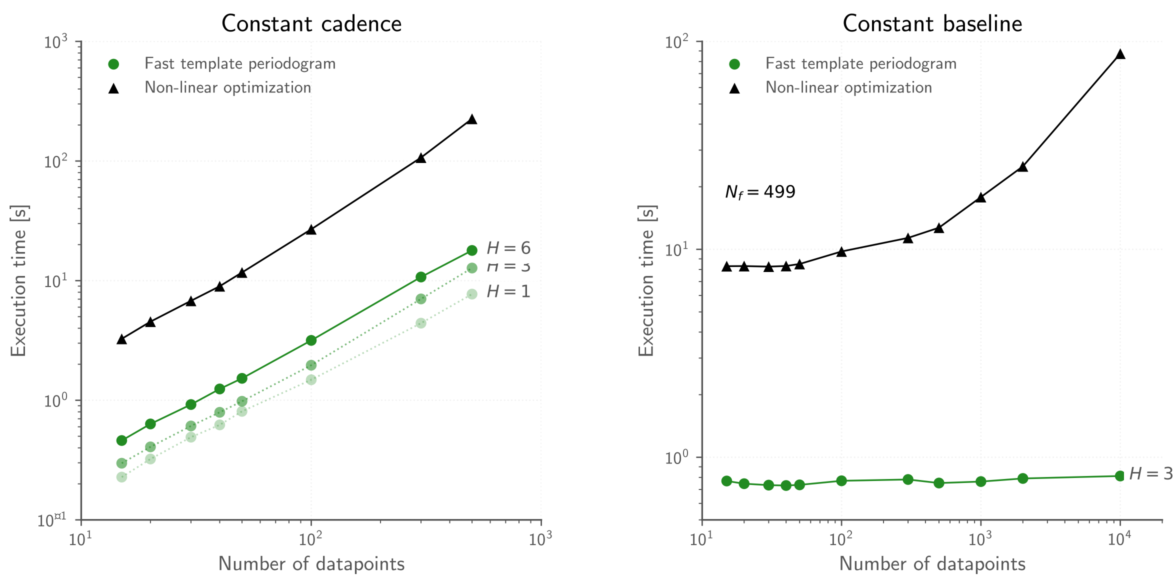

FIG. 6.— Computation time of FTP compared with alternative techniques that use non-linear optimization at each trial frequency. *Left*: timing for the case when $N_f = 12N_{\rm obs}$, i.e. the cadence of the observations is constant. *Right*: timing for the case when N_f is fixed, i.e. the baseline of the observations are constant. Non-linear optimization techniques scale as $\mathcal{O}(N_f N_{\rm obs})$ while the FTP scales as $\mathcal{O}(HN_f \log HN_f + N_f H^3)$, where H is the number of harmonics needed to approximate the template.

improve speedups by 1-2 orders of magnitude over non-linear optimization.

4. DISCUSSION

Template fitting is a powerful technique for accurately recovering the period and amplitude of objects with *a priori* known lightcurve shapes. It has been used in the literature by, e.g. Stringer et al. (2019); Sesar et al. (2016, 2010), to detect RR Lyrae in the SDSS, PS1, and DES datasets, where it has been shown to produce purer samples of RR Lyrae at a given completeness. The computational cost of current template fitting algorithms, however, prevents their application to larger datasets.

We have presented a novel template fitting algorithm that extends the Lomb-Scargle periodogram (Lomb 1976; Scargle 1982; Barning 1963; Vaníček 1971) to handle non-sinusoidal signals that can be expressed in terms of a fixed truncated Fourier series with a reasonably small number of harmonics ($H \lesssim 10$).

The fast template periodogram (FTP) asymptotically scales as $\mathcal{O}(N_f H^2 \log N_f H^2 + N_f H^4)$, while previous template fitting algorithms such as the one used in the `gatspy` library (VanderPlas 2016), scale as $\mathcal{O}(N_f N_{\rm obs})$. However, the FTP effectively scales as $\mathcal{O}(N_f H^4)$, since the time needed to compute polynomial coefficients and perform zero-finding dominates the computational time for all practical cases ($N_f \lesssim 10^{120}$). The H^4 scaling effectively restricts templates to those that are sufficiently smooth to be explained by a small number of Fourier terms.

FTP also improves the accuracy of previous template fitting algorithms, which rely on non-linear optimization at each trial frequency to minimize the V of the template fit. The FTP routinely finds superior fits over non-linear optimization methods.

An open-source Python implementation of the FTP is available at GitHub.[8] The current implementation could likely be improved by:

[8] https://github.com/PrincetonUniversity/FastTemplatePeriodogram

1. Improving the speed of the polynomial coefficient calculations and the zero-finding steps. This could potentially yield a speedup of $\sim 1-2$ orders of magnitude over the current implementation.

2. Exploiting the embarassingly parallel nature of the FTP using GPU's.

For a constant baseline, the current implementation improves existing methods by factors of a ~few for lightcurves with $\mathcal{O}(100)$ observations, and by an order of magnitude or more for objects with more than 1,000 observations. While template fitting remains challenging for the largest time-domain surveys at present, the polynomial computations themselves admit a further factor of $\sim$ 25–100 speedup, which would bring the FTP to 1–3 orders of magnitude faster than alternative techniques. This work presents the mathematical shortcut that, combined with a more efficient implementation, will make template fitting a fast and valuable tool for detecting non-sinusoidal periodic signals of specified shape in large time-series datasets.

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APPENDIX

FULL DERIVATION OF THE FAST TEMPLATE PERIODOGRAM

This appendix gives the explicit algebra summarised in Section 2. We present (i) the general form of the gradient conditions for a weighted-least-squares periodogram; (ii) the polynomial expansion of the phase-shifted template; (iii) the closed-form solution to the linear least-squares sub-problem at fixed phase shift θ_2; (iv) the reduction of the residual sum of squares V to the compact periodogram form of Equation 20; (v) the polynomial expansion of the weighted statistics MM and YM in terms of $\psi = e^{i\theta_2}$ and the trigonometric sums of Equations 23–27; and (vi) the algebraic identity converting the non-linear condition Equation 21 into the polynomial root-finding condition of Equation 22.

General form of the gradient conditions

The optimality condition Equation 17 can be written explicitly for each parameter θ_j as follows. Writing $\hat{y}_i = \hat{y}(\omega t_i;\theta)$ and $\partial_j \hat{y}_i = \partial \hat{y}(\omega t;\theta)/\partial\theta_j|_{t=t_i}$,

$$\begin{aligned} 0 &= \left.\frac{\partial V}{\partial \theta_j}\right|_{\theta=\theta_{\rm opt}} \\ &= -2\sum_i w_i\left(y_i - \hat{y}_i\right)(\partial_j\hat{y})_i, \\ \sum_i w_i y_i (\partial_j \hat{y})_i &= \sum_i w_i \hat{y}_i (\partial_j\hat{y})_i. \end{aligned} \tag{A1}$$

This is a general result that extends to all least-squares periodograms. To simplify the derivations below we adopt the weighted-average notation

$$\langle X\rangle \equiv \sum_i w_i X_i, \tag{A2}$$

$$\langle XY\rangle \equiv \sum_i w_i X_i Y_i, \tag{A3}$$

$$\mathrm{Cov}(X,Y) \equiv \langle XY\rangle - \langle X\rangle\langle Y\rangle, \tag{A4}$$

$$\mathrm{Var}(X) \equiv \mathrm{Cov}(X,X). \tag{A5}$$

Phase-shifted template expansion

The transposed template $\mathbf{M}_{\theta_2} = \mathbf{M}(\omega t - \theta_2)$ can be expressed as

$$\begin{aligned} \mathbf{M}_{\theta_2}(\omega t) &= \sum_n c_n \cos n\left(\omega t - \theta_2\right) + s_n \sin n\left(\omega t-\theta_2\right) \\ &= \sum_n \left(c_n\cos n\theta_2 - s_n \sin n\theta_2\right)\cos n\omega t \\ &\qquad + \left(s_n \cos n\theta_2 + c_n \sin n\theta_2\right)\sin n\omega t \\ &= \sum_n \left(\alpha_n\psi^n + \alpha_n^*\psi^{-n}\right)\cos n\omega t \\ &\qquad - i\left(\alpha_n\psi^n - \alpha_n^*\psi^{-n}\right)\sin n\omega t \\ &= \sum_n A_n(\psi)\cos n\omega t + B_n(\psi)\sin n\omega t, \end{aligned} \tag{A6}$$

where $\alpha_n = (c_n + i s_n)/2$ and $\psi \equiv e^{i\theta_2}$. We also define the weighted statistics

$$\begin{aligned} \widehat{YM} &\equiv \langle y\mathbf{M}_{\theta_2}\rangle, & \widehat{MM} &\equiv \langle \mathbf{M}_{\theta_2}^2\rangle, \\ \overline{M} &\equiv \langle \mathbf{M}_{\theta_2}\rangle, & MM &\equiv \mathrm{Var}(\mathbf{M}_{\theta_2}) = \widehat{MM} - \overline{M}^2, \\ YM &\equiv \mathrm{Cov}(\mathbf{M}_{\theta_2}, y) = \widehat{YM} - \bar{y}\overline{M}. \end{aligned} \tag{A7}$$

Closed-form linear least squares for (θ_1, θ_3)

For a given phase shift θ_2, the optimal amplitude and offset are obtained by setting the partial derivatives of V with respect to θ_1 and θ_3 to zero:

$$\begin{aligned} 0 = \frac{\partial V}{\partial\theta_1} &= -2\sum_i w_i(y_i - \theta_1\mathbf{M}_{\theta_2} - \theta_3)\mathbf{M}_{\theta_2} \\ &\propto \widehat{YM} - \theta_1\widehat{MM} - \theta_3\overline{M}, \end{aligned} \tag{A8}$$

$$\begin{aligned} 0 = \frac{\partial V}{\partial\theta_3} &= -2\sum_i w_i(y_i - \theta_1\mathbf{M}_{\theta_2} - \theta_3) \\ &\propto \bar{y} - \theta_1\overline{M} - \theta_3. \end{aligned} \tag{A9}$$

This 2×2 system can be written as

$$\begin{pmatrix} \widehat{MM} & \overline{M} \\ \overline{M} & 1 \end{pmatrix}\begin{pmatrix}\theta_1 \\ \theta_3\end{pmatrix} = \begin{pmatrix}\widehat{YM} \\ \bar{y}\end{pmatrix}, \tag{A10}$$

with closed-form solution

$$\begin{pmatrix}\theta_1\\ \theta_3\end{pmatrix} = \begin{pmatrix} YM/MM \\ \bar{y} - \overline{M}\,(YM/MM)\end{pmatrix}, \tag{A11}$$

where $MM = \widehat{MM} - \overline{M}^2$ and $YM = \widehat{YM} - \bar{y}\overline{M}$. The fitted model is then

$$\hat{y}_i = \bar{y} + \left(\frac{YM}{MM}\right)\left(M_i - \overline{M}\right). \tag{A12}$$

The periodogram in terms of YY, YM, and MM

Substituting Equation A11 into V,

$$\begin{aligned} V &= \sum_i w_i (y_i - \hat{y}_i)^2 = \sum_i w_i (y_i^2 - 2y_i\hat{y}_i + \hat{y}_i^2) \\ &= YY - 2\frac{(YM)^2}{MM} + \frac{(YM)^2}{MM} \\ &= YY - \frac{(YM)^2}{MM}. \end{aligned} \tag{A13}$$

Since $V_0 = YY$,

$$P(\omega) = 1 - \frac{V}{V_0} = \frac{(YM)^2}{YY \cdot MM}, \tag{A14}$$

which is Equation 20 of the body.

Polynomial expansion of MM and YM

The autocovariance of the template values, MM, is given by

$$\begin{aligned} MM &\equiv \sum_i w_i M_i^2 - \left(\sum_i w_i M_i\right)^2 \\ &= \sum_i w_i \left(\sum_n A_n \cos\omega n t_i + B_n \sin\omega n t_i\right)^2 \\ &\quad - \left(\sum_n A_n C_n + B_n S_n\right)^2 \\ &= \sum_{n,m} A_n A_m CC_{nm} + 2A_n B_m CS_{nm} \\ &\quad + B_n B_m SS_{nm}, \end{aligned} \tag{A15}$$

using the symmetry $\sum_{n,m} A_n B_m CS_{nm} = \sum_{n,m} A_m B_n CS_{mn}$.

The products $A_n A_m$, $A_n B_m$, $B_n B_m$ expand as

$$\begin{aligned} A_n A_m &= \alpha_n \alpha_m \psi^{n+m} + \alpha_n^* \alpha_m \psi^{m-n} \\ &\quad + \alpha_n \alpha_m^* \psi^{n-m} + \alpha_n^* \alpha_m^* \psi^{-n-m}, \end{aligned} \tag{A16}$$

$$\begin{aligned} A_n B_m &= -i\big[\alpha_n \alpha_m \psi^{n+m} + \alpha_n^* \alpha_m \psi^{m-n} \\ &\quad - \alpha_n \alpha_m^* \psi^{n-m} - \alpha_n^* \alpha_m^* \psi^{-n-m}\big], \end{aligned} \tag{A17}$$

$$\begin{aligned} B_n B_m &= -\big[\alpha_n \alpha_m \psi^{n+m} - \alpha_n^* \alpha_m \psi^{m-n} \\ &\quad - \alpha_n \alpha_m^* \psi^{n-m} + \alpha_n^* \alpha_m^* \psi^{-n-m}\big]. \end{aligned} \tag{A18}$$

Substituting these into the MM sum gives the polynomial form

$$\begin{aligned} MM_{nm} &= A_n A_m CC_{nm} + 2A_n B_m CS_{nm} + B_n B_m SS_{nm} \\ &= \alpha_n \alpha_m \widetilde{CC}_{nm} \psi^{n+m} + 2\alpha_n \alpha_m^* \widetilde{CS}_{nm} \psi^{n-m} \\ &\quad + \alpha_n^* \alpha_m^* \widetilde{SS}_{nm} \psi^{-(n+m)}, \end{aligned} \tag{A19}$$

where

$$\widetilde{CC}_{nm} = (CC_{nm} - SS_{nm}) - i\left(CS + CS^T\right)_{nm}, \tag{A20}$$

$$\widetilde{CS}_{nm} = (CC_{nm} + SS_{nm}) + i\left(CS - CS^T\right)_{nm}, \tag{A21}$$

$$\widetilde{SS}_{nm} = (CC_{nm} - SS_{nm}) + i\left(CS + CS^T\right)_{nm}. \tag{A22}$$

Analogously, the cross term YM expands as

$$\begin{aligned} YM_k &= A_k YC_k + B_k YS_k \\ &= \alpha_k YC_k \psi^k + \alpha_k^* YC_k \psi^{-k} \\ &\quad - i\left(\alpha_k YS_k \psi^k - \alpha_k^* YS_k \psi^{-k}\right) \\ &= \alpha_k \widetilde{YC}_k \psi^k + \alpha_k^* \widetilde{YC}_k^* \psi^{-k}, \end{aligned} \tag{A23}$$

where $\widetilde{YC}_k \equiv YC_k - iYS_k$.

From the non-linear condition to the polynomial form

The non-linear condition Equation 21 can be rewritten in terms of the rescaled quantities $YM' \equiv \psi^H YM$ and $MM' \equiv \psi^{2H} MM$, which (from Equations A19–A23) are polynomials in ψ of degree $2H$ and $4H$, respectively. Their derivatives are

$$\partial YM' = H\psi^{H-1} YM + \psi^H \partial YM, \tag{A24}$$

$$\partial MM' = 2H\psi^{2H-1} MM + \psi^{2H} \partial MM. \tag{A25}$$

Combining these,

$$\begin{aligned} 0 &= 2MM\,\partial(YM) - YM\,\partial(MM) \\ &= \psi^{3H+1}\left[2MM\,\partial(YM) - YM\,\partial(MM)\right] \\ &= 2\,\psi^{2H} MM\left[\psi^{H+1}\partial(YM)\right] \\ &\quad - \psi^H YM\left[\psi^{2H+1}\partial(MM)\right] \\ &= 2\,MM'\left[\psi\,\partial(YM') - H\,(YM')\right] \\ &\quad - YM'\left[\psi\,\partial(MM') - 2H\,(MM')\right] \\ &= \psi\left[2\,MM'\,\partial(YM') - YM'\,\partial(MM')\right] \\ &\quad - 2H\left[MM'\,YM' - YM'\,MM'\right] \\ &= 2\,MM'\,\partial(YM') - YM'\,\partial(MM'). \end{aligned} \tag{A26}$$

The last step assumes $\psi \neq 0$, which holds since $\psi = e^{i\theta_2}$ lies on the unit circle for all real θ_2. Equation A26 is the polynomial root-finding condition of Equation 22 in the body.

SHARED-PHASE MULTI-BAND TEMPLATE PERIODOGRAM

We derive a multi-band extension for the template periodogram for data taken in K filters, with N_k observations in the k-th filter. We use the same model as the one described in Sesar et al. (2016) in order to illustrate the applicability of the template periodogram to more sophisticated scenarios.

The i-th observation in the k-th filter is denoted $y_i^{(k)}$. We wish to fit a periodic, multi-band template $\mathbf{M} = (\mathbf{M}^{(1)}, \mathbf{M}^{(2)}, ..., \mathbf{M}^{(K)}) : [0, 2\pi) \to \mathbb{R}^K$ to all observations. We assume the same model used by Sesar et al. (2016), which assumes the relative amplitudes, phase shifts, and offsets are shared across bands:

$$\hat{y}^{(k)}(t;\theta) = \theta_1 \mathbf{M}^{(k)}(\omega t - \theta_2) + \theta_3 + \lambda^{(k)} \tag{B1}$$

where $\lambda^{(k)}$ is a fixed relative offset for band k. The V for this model is

$$V = \sum_{k=1}^{K} \sum_{i=1}^{N_k} w_i^{(k)} \left(y_i^{(k)} - \hat{y}_i^{(k)}\right)^2 \tag{B2}$$

To make things simpler, we can set the $\lambda^{(k)}$ values to 0 simply by subtracting them off from our observations; this means we take all $y_i^{(k)} \to y_i^{(k)} - \lambda^{(k)}$. We have that

$$V = \sum_{k=1}^{K} W^{(k)} V_k \tag{B3}$$

Where $W^{(k)} \equiv \sum_{i=1}^{N_k} w_i^{(k)}$ and $\sum_{k=1}^{K} W^{(k)} = 1$. This means that the system of equations reduces to:

$$0 = \frac{\partial V}{\partial \theta_1} = \sum_{k=1}^{K} W^{(k)} \left(\widehat{YM}^{(k)} - \theta_1 \widehat{MM}^{(k)} - \theta_3 \overline{M}^{(k)} \right) \tag{B4}$$

for the θ_1 parameter, where $\widehat{YM}^{(k)}$, $\widehat{MM}^{(k)}$, $\overline{M}^{(k)}$ are values from the single band case computed for each band individually, holding $W^{(k)} = 1$ for each band. That is:

$$\widehat{YM}^{(k)} \equiv \frac{1}{W^{(k)}} \sum_{i=1}^{N_k} w_i^{(k)} y_i^{(k)} \mathbf{M}_{\theta_2}^{(k)}(\omega t_i) \tag{B5}$$

$$\widehat{MM}^{(k)} \equiv \frac{1}{W^{(k)}} \sum_{i=1}^{N_k} w_i^{(k)} \left(\mathbf{M}_{\theta_2}^{(k)}(\omega t_i) \right)^2 \tag{B6}$$

$$\overline{M}^{(k)} \equiv \frac{1}{W^{(k)}} \sum_{i=1}^{N_k} w_i^{(k)} \mathbf{M}_{\theta_2}^{(k)}(\omega t_i) \tag{B7}$$

For the offset θ_3, we have

$$0 = \frac{\partial V}{\partial \theta_3} = \sum_{k=1}^{K} W^{(k)} \left(\bar{y}^{(k)} - \theta_1 \overline{M}^{(k)} - \theta_3 \right) \tag{B8}$$

Where $\bar{y}^{(k)}$ is the weighted-mean for the k-th band ($\bar{y}^{(k)} \equiv (1/W^{(k)}) \sum_{i=1}^{N_k} w_i^{(k)} y_i^{(k)}$), again with $y_i^{(k)} \to y_i^{(k)} - \lambda^{(k)}$. So if we redefine the quantities $\widehat{YM}$, $\widehat{MM}$, $\overline{M}$, and $\bar{y}$ as weighted averages across the bands, i.e. $\widehat{YM} = \sum_{k=1}^{K} W^{(k)} \widehat{YM}^{(k)}$, etc., the solution for the optimal parameters has the same form:

$$\theta_1 = \left(\frac{YM}{MM} \right) \tag{B9}$$

$$\theta_3 = \bar{y} - \overline{M} \left(\frac{YM}{MM} \right) \tag{B10}$$

which means that the model for the k-th band is

$$\hat{y}^{(k)} = \bar{y} + \left(\frac{YM}{MM} \right) \left(M_i^{(k)} - \overline{M} \right) \tag{B11}$$

The form of the periodogram has the same form as the single-band case:

$$\begin{aligned}
V &= \sum_{k=1}^{K} \sum_{i=1}^{N_k} w_i^{(k)} \left(y_i^{(k)} - \hat{y}_i^{(k)} \right)^2 && \text{(B12)} \\
&= \sum_{k=1}^{K} \sum_{i=1}^{N_k} w_i^{(k)} \left((y_i^{(k)} - \bar{y}) - \left(\frac{YM}{MM} \right) (M_i^{(k)} - \overline{M}) \right)^2 && \text{(B13)} \\
&= \sum_{k=1}^{K} \sum_{i=1}^{N_k} w_i^{(k)} \left((y_i^{(k)} - \bar{y})^2 - 2 \left(\frac{YM}{MM} \right) (M_i^{(k)} - \overline{M})(y_i^{(k)} - \bar{y}) + \left(\frac{YM}{MM} \right)^2 (M_i^{(k)} - \overline{M})^2 \right)^2 && \text{(B14)} \\
&= YY - 2\frac{(YM)^2}{MM} + \frac{(YM)^2}{MM} && \text{(B15)} \\
&= YY - \frac{(YM)^2}{MM}. && \text{(B16)}
\end{aligned}$$

Since there is a single shared offset between the bands (i.e. we assume the mean magnitude is the same in all bands after subtracting $\lambda^{(k)}$), the variance for the signal, YY, is not $\sum_{k=1}^{K}\sum_{i=1}^{N_k} w_i^{(k)}(y_i^{(k)} - \bar{y}^{(k)})^2$ but $\sum_{k=1}^{K}\sum_{i=1}^{N_k} w_i^{(k)}(y_i^{(k)} - \bar{y})^2$.

Construction of the polynomial for the multi-band case can be performed by first computing the polynomial expression for $\psi^{2H}\widehat{MM} = \sum_{k=1}^{K}\sum_{i=1}^{N_k} w_i^{(k)}\left(\psi^H M_i^{(k)}\right)^2$ and a separate polynomial expression for $\psi^H\overline{M}$, which can then be squared and subtracted from $\psi^{2H}\widehat{MM}$ to find $MM' = \psi^{2H}MM$.

For $YM' = \psi^H YM$, we merely compute $\psi^H\widehat{YM}^{(k)}$ for each band, take the weighted average of the polynomial coefficients for all bands ($\psi^H\widehat{YM} = \sum_k^K W^{(k)}\psi^H\widehat{YM}^{(k)}$) and subtract the $\psi^H\bar{y}\overline{M}$ polynomial to get YM'.

After computing the polynomials MM' and YM', the polynomial in Equation 22 can be computed quickly and the following steps for finding the optimal model parameters is the same as in the single band case.

The computational complexity of this model scales as $\mathcal{O}(KN_fH\log N_fH + N_fH^3)$ where K is the number of filters. Since the polynomial zero-finding step is the limiting computation in most real-world applications, the multi-band template periodogram corresponding to the Sesar et al. (2016) model should not be significantly more computationally intensive than the single-band case.

DATA AVAILABILITY

The data underlying this article are available in the GitHub repository located at https://github.com/PrincetonUniversity/FastTemplatePeriodogramPaper.